

# **DAYANANDA SAGAR COLLEGE OF ENGINEERING**

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning**



## **INTERNSHIP REPORT**

**ON**

### **“Machine Learning Intern at CodTech IT Solutions Pvt Ltd”**

*Submitted in partial fulfillment for the award of degree(20INT82)*

## **BACHELOR OF ENGINEERING IN**

**Department of Artificial Intelligence and Machine Learning**

*Submitted by:*

**SK Sai Tarun - 1DS21AI050**

*Conducted at*



**CODTECH IT SOLUTIONS PVT.LTD**

**Department of Artificial Intelligence and Machine Learning  
Dayananda Sagar College of Engineering Bangalore-78**

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



## CERTIFICATE

This is to certify that the Internship titled “**Machine Learning Intern at CodTech IT Solutions Pvt Ltd**” carried out by **Mr. SK SAI TARUN [1DS21AI050]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

**Signature of Reporting Head**  
**Neela Santhosh Kumar**  
HR  
CodTech IT  
Solutions Pvt  
Ltd

**Signature of Guide**  
**Prof. Ensteih Silvia**  
Dept. of AI & ML  
DSCE

**Signature of HoD**  
**Dr. Vindhya P Malagi**  
Dept. of AI & ML  
DSCE

### External Viva:

Name of the Examiner

1) \_\_\_\_\_

2) \_\_\_\_\_

Signature with Date

\_\_\_\_\_

\_\_\_\_\_

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



## DECLARATION

I, **SK Sai Tarun**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **CODTECH IT SOLUTIONS PVT.LTD**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI050

Place: Bangalore

NAME: SK Sai Tarun

# OFFER LETTER



## **CODTECH IT SOLUTIONS PVT.LTD** IT SERVICES & IT CONSULTING

8-7-7/2, Plot NO.51, Opp: Naveena School, Hasthinapuram Central, Hyderabad , 500 079. Telangana

### **Internship Offer Letter**



Date : 08/02/2025

Intern ID :CT4MTDA

Dear SK Sai Tarun

Congratulations on being selected for the **Machine Learning**. We at **CODTECH IT SOLUTIONS PVT.LTD** are thrilled to have you join our team. **This Online Internship will span 4 months from February 10th, 2025 to June 10th, 2025.**

This internship is designed as an educational experience, focusing on learning, skill development, and gaining practical knowledge. As an intern, we expect you to:

1. Complete all assignments to the best of your ability.
2. Follow any lawful and reasonable instructions provided by your supervisors.
3. Participate actively in team meetings and discussions.
4. Provide regular updates on your progress.
5. Adhere to company policies and maintain a professional demeanor.
6. Collaborate effectively with team members and contribute to group projects.
7. Seek feedback and apply it to improve your performance.

We trust that you will approach all tasks with diligence and enthusiasm. We are confident that this internship will be an enriching experience for you. We look forward to working with you and supporting you in achieving your career aspirations

Best regards,

Neela Santhosh Kumar

Human Resources & Academic Head



VERIFIED BY



CODTECH IT SOLUTIONS PRIVATE LIMITED CODTECH IT SOLUTIONS PVT LTD

[www.codtechitsolutions.com](http://www.codtechitsolutions.com)



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Hr@codtechitsolutions.com

## ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to the Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing me an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Mr. Neela Santhosh Kumar**, HR&AH, CodTech IT Solutions Pvt Ltd, for guiding me during the period of internship.

I express my deep and profound gratitude to my Guide, **Prof. Ensteih Silvia**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible.

**SK Sai Tarun[1DS21AI050]**

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# CHAPTER 1

## INTRODUCTION

### INTRODUCTION TO INTERNSHIP WORK

In the rapidly evolving field of retrieval-augmented generation (RAG) and large language model (LLM) fine-tuning, embedding model selection plays a crucial role in determining the effectiveness of information retrieval systems. Embeddings transform textual data into numerical representations, enabling models to efficiently retrieve relevant information from large knowledge bases. The comparison of different embedding models helps in identifying the most suitable approach for improving retrieval accuracy and downstream tasks in RAG-based applications.

Fine-tuning LLMs like LLaMA 3 is another essential step in customizing models for domain-specific tasks. Fine-tuning allows pre-trained models to adapt to specialized datasets, improving their understanding and generation of relevant content. Some key benefits of embedding model comparison and fine-tuning include:

1. **Optimized Retrieval Performance:** Evaluating various embedding models ensures that the most effective one is selected for accurate and efficient document retrieval in RAG systems.
2. **Improved Contextual Understanding:** Fine-tuning LLaMA 3 on domain-specific data enhances its ability to generate relevant and coherent responses.
3. **Scalability and Efficiency:** Selecting the right embeddings and fine-tuning LLMs contribute to faster and more resource-efficient inference in large-scale applications.
4. **Enhanced Personalization:** Fine-tuned models perform better in specialized domains, enabling better customization for industry-specific tasks.
5. **Data-Driven Insights:** Through model evaluation and tuning, organizations can derive actionable insights to refine their AI-powered solutions.

In conclusion, embedding model comparison and LLM fine-tuning are critical for optimizing RAG-based applications. By systematically evaluating embedding models and leveraging fine-tuned LLMs, organizations can enhance information retrieval accuracy, improve language generation, and develop more effective AI-driven systems.

## **CHAPTER 2**

### **DESCRIPTION OF ORGANIZATION**

#### **CODTECH IT SOLUTIONS PVT LTD**



#### **ABOUT THE COMAPNY**

CodTech IT Solutions Pvt Ltd is a dynamic Indian information technology company established in the early 2020s. Headquartered in Hyderabad, Telangana, the company operates as a private limited entity and serves a wide range of clients including startups, small businesses, and educational institutions. Founded by Mr. Neelam Harish, who serves as the Chief Executive Officer, and Mr. Neela Akhil Kumar, who functions as the Technical Head and Co-founder, CodTech has rapidly gained recognition for offering end-to-end IT services. The company operates across various technology domains including software development, web and mobile application creation, cybersecurity, cloud solutions, and artificial intelligence.

The organization has positioned itself as a one-stop digital partner for businesses and entrepreneurs seeking to expand or modernize their operations through technology. With a skilled workforce and a client-first approach, CodTech blends technological expertise with market understanding to deliver scalable, secure, and innovative solutions. In addition to its commercial services, CodTech also runs training and internship programs that are designed to equip students and fresh graduates with hands-on technical skills in trending areas such as full-stack development, data science, and machine learning.

#### **VISION:**

The vision of CodTech IT Solutions Pvt Ltd is to become the most trusted technology partner for small businesses across various sectors. The company aspires to empower its clients with innovative, secure, and scalable solutions that address their unique business challenges.



By focusing on advanced technologies such as web and app development, artificial intelligence, data science, and cybersecurity, CodTech seeks to play a transformative role in the growth and digitalization of its clients' operations.

The company envisions a future where businesses of all sizes have access to world-class IT solutions, regardless of budget or complexity. It aims to drive this vision by building reliable systems, enhancing digital engagement, and ensuring secure technology infrastructures for its clients. CodTech believes that by supporting small businesses with tailored technology, it can contribute to larger economic development and technological progress.

### **MISSION:**

CodTech's mission revolves around creating meaningful value through technology. The company is dedicated to empowering businesses by providing custom-designed digital solutions that are efficient, affordable, and future-ready. Although it does not formally outline a specific mission statement, the organization's consistent messaging and service offerings convey a clear purpose: to enable growth, operational efficiency, and digital presence for its clients through a combination of technical innovation and strategic support.

In addition to delivering business solutions, CodTech maintains a strong educational mission. The company believes in nurturing the next generation of technology professionals by offering structured internship programs, practical training, and certification opportunities. These initiatives are designed to bridge the skill gap in the IT industry while supporting internal talent pipelines. CodTech upholds the belief that its clients' success is directly tied to its own and reflects this in its tagline: "Your success is our mission."

### **CORE AREAS OF INNOVATION:**

CodTech IT Solutions Pvt Ltd engages in a wide spectrum of technology-driven services, with a focus on continual innovation. The core areas where the company is driving change and adding value include:

- **Web and Mobile Application Development:**

CodTech builds secure, responsive, and scalable websites and mobile apps for industries such as healthcare, education, retail, and services.

- **Artificial Intelligence and Automation:**

The company develops AI-powered tools and automation workflows to increase efficiency, reduce costs, and enhance decision-making for clients.

- **Data Science and Analytics:**

CodTech provides predictive analytics, big data solutions, and custom dashboards to help businesses extract actionable insights from data.

- **Cybersecurity:**

The company offers robust cybersecurity services including penetration testing, secure application development, and compliance solutions to protect digital assets.

- **Cloud Computing:**

CodTech supports cloud migration, infrastructure deployment, and maintenance using platforms such as AWS and Microsoft Azure, ensuring flexibility and scalability.

- **Digital Marketing and Branding:**

The company delivers digital marketing strategies including SEO, social media management, and pay-per-click advertising to improve online visibility.

- **Training and Internship Programs:**

CodTech runs practical, project-based internship programs in full-stack development, machine learning, Python, Java, .NET, and more.

- **Industry-Specific Platforms:**

Tailored software systems are built for industries like healthcare and education, including hospital portals and learning management systems.

## CHAPTER 3

### SCOPE OF WORK

During the course of my internship, I was involved in two core projects that explored cutting-edge applications of Large Language Models (LLMs) and retrieval-augmented generation (RAG) in the domain of Natural Language Processing (NLP). The work provided deep insights into language model fine-tuning, evaluation of embedding models, and optimizing LLMs for domain-specific tasks.

#### 1. Embedding Model Comparison for RAG:

Objective: The objective of this project was to compare and evaluate the performance of different embedding models for their effectiveness in a Retrieval-Augmented Generation (RAG) pipeline.

Scope: The scope involved -

- Explored Cohere and Voyage embedding models in the context of financial question answering.
- Constructed a RAG pipeline where retrieved context was combined with user queries to generate responses.
- Evaluated generated outputs using metrics such as Exact Match (EM), F1 Score, BLEU, and ROUGE.
- Analyzed the trade-offs between embedding accuracy, contextual relevance, and downstream generation quality.

Methodology: Generating embeddings using Cohere and Voyage-02 models and evaluated in a RAG pipeline using financial QA data. Top-k context retrieval was performed via cosine similarity, and results were scored using EM, F1, BLEU, and ROUGE metrics.

Results: The project provided insights into how embedding model choice impacts the quality of retrieval and answer generation in RAG systems. Though performance was similar across models, metric-based comparison revealed subtle differences, guiding future embedding selection for domain-specific tasks.

## 2. Finetuning Llama-3

Objective: This project aimed to fine-tune the Meta LLaMA-3 8B Instruct model for improved performance on a financial question answering dataset.

Scope:

- Utilized a dataset containing 10,000 financial questions, contexts, and answers.
- Pre-processed and formatted the data using chat templates compatible with LLaMA-3.
- Employed parameter-efficient fine-tuning using LoRA (Low-Rank Adaptation).
- Trained the model using SFTTrainer with quantized 4-bit weights to reduce computational load.
- Deployed and evaluated the fine-tuned model using generated predictions and standard NLP metrics.

Methodology: The LLaMA 3 (8B Instruct) model was fine-tuned using LoRA and SFTTrainer on formatted financial QA pairs. Quantization was applied for efficiency, and the model's outputs were evaluated on a held-out test set.

Results: The fine-tuned model demonstrated significant improvements in generating contextually accurate answers for financial queries. The experience also enhanced my understanding of fine-tuning workflows, tokenization strategies, and efficient LLM deployment.

In conclusion, the internship projects enabled hands-on experience with advanced LLM techniques including retrieval-based generation, metric evaluation, and parameter-efficient fine-tuning. They offered exposure to state-of-the-art tools like Hugging Face Transformers, PEFT, and LLaMA 3 models, contributing to my technical growth in applied machine learning and NLP.

## **CHAPTER 4**

### **LEARNING OBJECTIVES**

#### **1. Understanding of Retrieval-Augmented Generation (RAG):**

One of the core learning objectives during the internship was to develop a comprehensive understanding of Retrieval-Augmented Generation (RAG) architectures and their real-world applications. This involved exploring the interaction between dense retrieval techniques and generative language models, particularly in open-domain question-answering tasks. By experimenting with multiple embedding models such as Cohere and Voyage-02, I gained exposure to vector representation of texts, semantic similarity, and information retrieval strategies. I also evaluated retrieval effectiveness using standardized metrics like Exact Match (EM), F1 Score, BLEU, and ROUGE, thereby learning how retrieval quality directly influences final answer generation in a RAG pipeline.

#### **2. Practical Skills in Large Language Model Fine-Tuning:**

Another major objective was to gain hands-on experience in fine-tuning state-of-the-art large language models, specifically the Meta LLaMA 3-8B Instruct model. I learned to use parameter-efficient fine-tuning techniques such as LoRA (Low-Rank Adaptation), enabling the adaptation of large-scale models using minimal computational resources. This involved working with Hugging Face's transformers, trl, and peft libraries, using 4-bit quantized models for memory-efficient training. Through this, I deepened my understanding of instruction tuning and prompt-based learning in transformer models.

#### **3. Development of Data Processing and Prompt Engineering Capabilities:**

A significant part of the project required transforming raw financial question-answer datasets into a structured format compatible with instruction-tuned models. I learned to engineer prompts following chat-based templates, ensuring clarity, context preservation, and alignment with system messages. The process of prompt engineering taught me how model inputs influence the generated outputs, and how structured formatting can improve the accuracy and relevance of model predictions.

#### **4. Mastery of Model Evaluation and Benchmarking Techniques:**

To effectively compare the performance of both embedding and language models, I developed skills in applying various evaluation metrics across retrieval and generation stages. I interpreted the implications of metrics like BLEU and ROUGE for generative quality and EM and F1 for retrieval accuracy. These skills allowed me to identify trade-offs between different model choices, diagnose failure cases, and iterate model configurations based on empirical results.

#### **5. Understanding of End-to-End Machine Learning Workflows:**

Throughout the internship, I engaged in the complete machine learning pipeline—from dataset creation and preprocessing to model training, evaluation, and deployment. I learned to manage experiments, track model versions, and utilize Hugging Face Hub for sharing and storing trained models. This experience helped me understand real-world challenges in reproducibility, scalability, and model sharing in modern AI development workflows.

#### **6 . Application of NLP and AI in Domain-Specific Contexts:**

The work was deeply rooted in applying AI and natural language processing (NLP) in the financial domain. This helped me understand the importance of domain-specific finetuning and the necessity of curating task-relevant data. It also highlighted how general-purpose LLMs can be customized for specialized knowledge tasks, such as financial QA, by aligning training data with expected model behaviour.

# CHAPTER 5

## CHALLENGES AND SOLUTION

### 1. Embedding Model Comparison for RAG:

#### Challenges:

- **Model Selection and Benchmarking:** One of the primary challenges was selecting appropriate embedding models that could effectively capture semantic similarity for document retrieval tasks. Each model had different characteristics (token limits, contextual understanding, inference cost), making it difficult to establish a fair baseline.
- **Evaluation Metrics:** Choosing meaningful metrics to evaluate retrieval quality was non-trivial. Standard retrieval metrics like Recall, MRR, and nDCG had to be interpreted in context, and customizing evaluation scripts for specific use cases took considerable effort.
- **Dataset Relevance:** Identifying or curating a dataset that realistically reflected downstream QA requirements posed a challenge. Off-the-shelf datasets didn't always align well with the retrieval goals, leading to noisy or misleading evaluation.
- **Integration with RAG Pipeline:** Integrating the embedding models into a full RAG pipeline and ensuring compatibility with retrievers (e.g., FAISS or Elasticsearch) required detailed understanding of embedding dimensions, normalization, and indexing strategies.

#### Solution:

To address these challenges, I began by selecting two prominent embedding models—one open-source and one commercial and developed a modular evaluation framework. This allowed side-by-side comparison across metrics like Recall@5 and cosine similarity. The evaluation pipeline was tested on a domain-relevant dataset, and qualitative assessment was also done to validate semantic matches. I implemented FAISS for vector search and ensured consistent preprocessing between indexing and query stages. Fine-tuning thresholds and vector normalization further improved retrieval quality. This work clarified the trade-offs between model performance, latency, and cost, and helped in recommending the most effective embedding model for downstream RAG tasks.

### 2. Fine-tuning LLaMA 3:

#### Challenges:

- **Model Infrastructure Requirements:** Fine-tuning large models like LLaMA 3 required high-performance compute infrastructure, especially GPUs with large memory. Access to such resources was limited, and ensuring stable training without resource bottlenecks was difficult.

- **Data Alignment and Quality:** Preparing data for fine-tuning involved aligning it with the intended task format. Ensuring high-quality, instruction-following data (e.g., JSONL format for instruction-tuning) required careful curation and filtering.
- **Hyperparameter Tuning:** Determining optimal hyperparameters—learning rate, batch size, weight decay, and number of training steps—was time-consuming. The wrong settings could lead to model overfitting, underfitting, or instability during training.
- **Monitoring and Debugging:** LLM fine-tuning is resource-intensive and time-consuming. Detecting issues mid-training (e.g., loss divergence, overfitting) and implementing early-stopping or checkpoints added complexity.

#### Solution:

To overcome these challenges, I utilized tools like Hugging Face's Transformers and trl libraries, specifically SFTTrainer, to streamline the fine-tuning process. I started with instruction-tuned checkpoints to reduce training cost and initialized training with low learning rates and gradient accumulation to adapt to GPU constraints. Datasets were cleaned and preprocessed using prompt formatting aligned with the Alpaca/LLaMA instruction styles. I implemented periodic evaluation on held-out validation sets and tracked training metrics to ensure model convergence. This led to a stable and task-adapted version of LLaMA 3 capable of better task-following.



## **HARDWARE AND SOFTWARE REQUIREMENTS**

### **Software requirements:**

- Operating system: Windows 10 64 bit
- Python: 3.10+
- PyTorch: 2.1.2 (with CUDA support)
- Transformers (Hugging Face): 4.39.3
- trl (for fine-tuning): 0.7.4
- Pinecone
- Cohere API
- Voyage API
- Jupyter Notebook / VS Code / Terminal-based workflow

### **Hardware Requirements:**

- Processor: Intel i9 / AMD Ryzen 9 (or equivalent high-performance CPU)
- RAM: Minimum 8 GB (for handling large datasets and model pipelines)
- GPU: NVIDIA A100 / V100 / RTX 3050 (Minimum 24 GB VRAM recommended for LLaMA 3 fine-tuning)
- Input Devices: Mouse, Keyboard
- LAN Connection Port: Required
- Internet Fiber Optic Cable: Required

# CHAPTER 6

## ACHIEVEMENTS AND CONTRIBUTION

### 1. Embedding Model Comparison for RAG:

#### Data Pre-processing and Analysis:

The project involved the development and evaluation of a Retrieval-Augmented Generation (RAG) pipeline by comparing two state-of-the-art embedding models: Cohere and Voyage. The objective was to determine which embedding model yields better retrieval performance for downstream question-answering tasks.

The dataset used consisted of domain-specific documents segmented into chunks for efficient semantic retrieval. Preprocessing involved:

- Cleaning and standardizing raw textual data.
- Splitting documents into overlapping chunks.
- Converting text to vector embeddings using Cohere and Voyage models.
- Indexing embeddings using FAISS for efficient nearest neighbour retrieval.

#### Modelling:

Each embedding model was integrated into a RAG pipeline using a fixed LLM decoder (e.g., GPT-based model). For each query:

- Top-k most relevant chunks were retrieved based on embedding similarity.
- Retrieved contexts were fed into the decoder to generate answers.

#### Evaluation metrics:

- Precision for document retrieval.
- BLEU, ROUGE, and F1-score for answer quality.

#### Key observations:

- Voyage embeddings outperformed Cohere in retrieval precision for domain-specific content.
- Cohere provided slightly faster embedding inference times.

## 2 .Fine-tuning of LLaMA 3 for Domain-Specific QA:

### Data Pre-processing and Analysis:

The fine-tuning of LLaMA 3 was conducted using a set of domain-specific question-answer pairs.

Steps included:

- Cleaning and deduplicating raw QA pairs.
- Formatting the data into instruction-tuning format (<instruction>, <input>, <output>).
- Tokenizing the dataset using LLaMA-compatible tokenizer.
- Splitting the dataset into training and validation sets (typically 90-10 split).

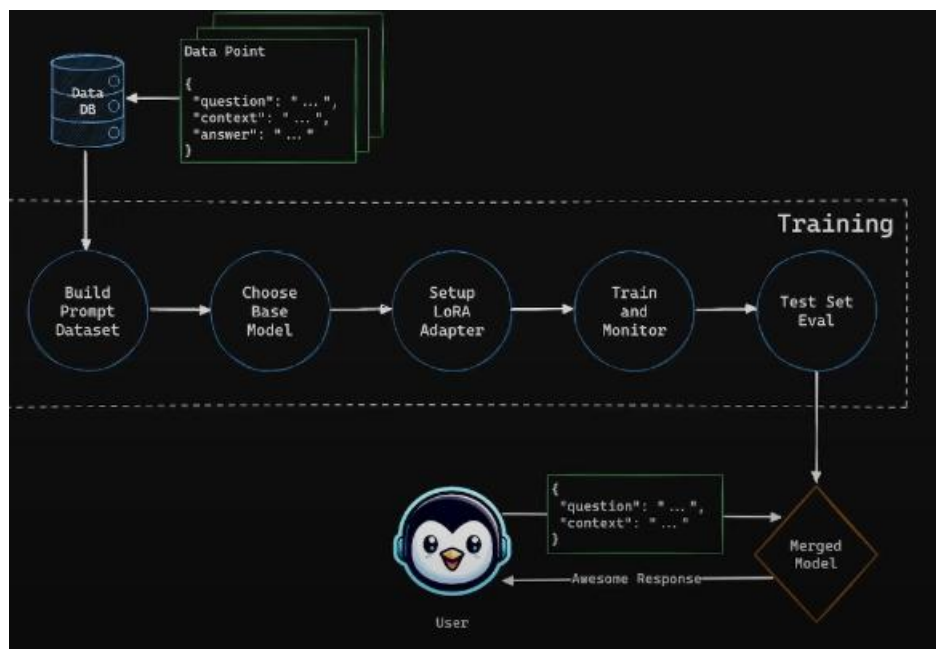


Fig 6.2: Finetuning process

| question<br>string · lengths                                                                        | answer<br>string · lengths                                                                                      | context<br>string · lengths                                                                                               |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 0 224                                                                                               | 0 736                                                                                                           | 0 5.59k                                                                                                                   |
| What area did NVIDIA initially focus on before expanding to other computationally intensive fields? | NVIDIA initially focused on PC graphics.                                                                        | Since our original focus on PC graphics, we have expanded to several other large and important computationally intensive. |
| What are some of the recent applications of GPU-powered deep learning as mentioned by NVIDIA?       | Recent applications of GPU-powered deep learning include recommendation systems, large language models, and...  | Some of the most recent applications of GPU-powered deep learning include recommendation systems, which are AI.           |
| What significant invention did NVIDIA create in 1999?                                               | NVIDIA invented the GPU in 1999.                                                                                | Our invention of the GPU in 1999 defined modern computer graphics and established NVIDIA as the leader in computer.       |
| How does NVIDIA's platform strategy contribute to the markets it serves?                            | NVIDIA's platform strategy brings together hardware, systems, software, algorithms, libraries, and services to. | NVIDIA has a platform strategy, bringing together hardware, systems, software, algorithms, libraries, and services to.    |

Fig.6.3 : Dataset

```

LlamaForCausalLM(
  (model): LlamaModel(
    (embed_tokens): Embedding(128264, 4096)
    (layers): ModuleList(
      (0-31): 32 x LlamaDecoderLayer(
        (self_attn): LlamaSdpaAttention(
          (q_proj): Linear4bit(in_features=4096, out_features=4096, bias=False)
          (k_proj): Linear4bit(in_features=4096, out_features=1024, bias=False)
          (v_proj): Linear4bit(in_features=4096, out_features=1024, bias=False)
          (o_proj): Linear4bit(in_features=4096, out_features=4096, bias=False)
          (rotary_emb): LlamaRotaryEmbedding()
        )
        (mlp): LlamaMLP(
          (gate_proj): Linear4bit(in_features=4096, out_features=14336, bias=False)
          (up_proj): Linear4bit(in_features=4096, out_features=14336, bias=False)
          (down_proj): Linear4bit(in_features=14336, out_features=4096, bias=False)
          (act_fn): SiLU()
        )
        (input_layernorm): LlamaRMSNorm()
        (post_attention_layernorm): LlamaRMSNorm()
      )
    )
    (norm): LlamaRMSNorm()
  )
  (lm_head): Linear(in_features=4096, out_features=128264, bias=False)

```

Fig.6.4 : LLM layers

```

trainable params: 83,886,000 || all params: 8,114,212,864 || trainable%: 1.0338

```

Fig.6.5 : No. of parameters

[500/500 1:45:16, Epoch 1/1]

| Step | Training Loss | Validation Loss |
|------|---------------|-----------------|
| 100  | 0.474600      | 0.454343        |
| 200  | 0.419800      | 0.445155        |
| 300  | 0.373500      | 0.436959        |
| 400  | 0.407200      | 0.440568        |
| 500  | 0.341400      | 0.433227        |

Fig.6.6 : Training

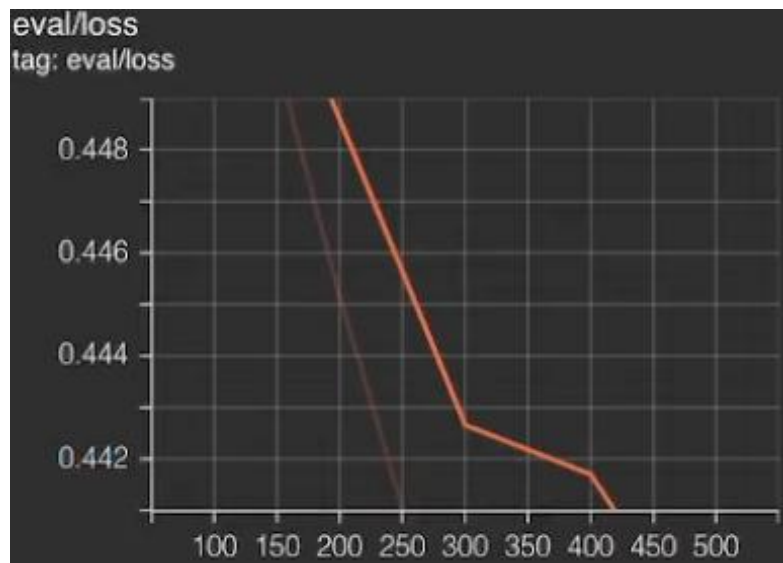


Fig.6.7 : Evaluation loss



Fig.6.8: Training loss

### Modelling:

The model selected was **LLaMA 3 8B**, fine-tuned using low-rank adaptation (LoRA) to reduce compute costs. Fine-tuning configuration:

- Optimizer: AdamW
- Learning Rate: 2e-5
- Epochs: 3
- Batch size: 8 (with gradient accumulation)
- Mixed precision training using FP16

Training was carried out using HuggingFace's Trainer API with LoRA adapters via peft library.

Evaluation:

- Evaluated on both held-out validation data and unseen real-world queries.
- Metrics: Perplexity, BLEU, ROUGE, and qualitative manual review.

Key outcomes:

- Fine-tuned LLaMA 3 significantly outperformed the base model on specific domain queries.
- Generated responses were more factual, structured, and aligned with domain vocabulary.

## CHAPTER 7

### REFLECTION AND ANALYSIS

During the course of this project, I developed a deeper understanding of modern Natural Language Processing (NLP) systems, particularly in the areas of retrieval-augmented generation (RAG) and large language model fine-tuning. My work centered around two core components: evaluating and comparing embedding models for information retrieval, and fine-tuning LLaMA 3 for domain-specific question answering.

Working on embedding model comparison introduced me to the intricacies of dense retrieval systems. I gained hands-on experience in preparing domain-specific corpora, implementing embedding-based retrieval pipelines, and rigorously benchmarking models using metrics like Recall, MRR, and nDCG. This process sharpened my understanding of how different sentence embedding models behave in real-world RAG applications, and how subtle variations in embedding quality can drastically impact downstream generation quality.

The fine-tuning of LLaMA 3 allowed me to explore the challenges and techniques involved in adapting a large language model to a specific task. I learned to work with instruction-style datasets, apply LoRA (Low-Rank Adaptation) for parameter-efficient fine-tuning, and evaluate model performance using custom QA benchmarks. This part of the project taught me how model performance can be enhanced through careful data curation and alignment strategies, especially in specialized domains.

Beyond the technical implementation, this experience gave me a practical framework for end-to-end LLM development — from retrieval to generation. I was able to appreciate the broader system design challenges in building effective RAG pipelines, such as latency-accuracy trade-offs, prompt engineering for context injection, and evaluating factual consistency.

Additionally, engaging in this work provided insight into current trends in LLM deployment and the increasing importance of domain adaptation in real-world AI applications. It emphasized the value of open-source tools, reproducible research practices, and interpretability in designing robust AI systems.

In conclusion, this project not only helped me develop concrete skills in retrieval, modeling, and evaluation, but also offered a strategic perspective on how to build intelligent systems that combine knowledge retrieval and generation effectively. These learnings have been pivotal in shaping my understanding of applied AI and will serve as a strong foundation for future work in NLP and generative AI systems.

## CHAPTER 8

### RECOMMENDATIONS

- 1. Expand Access to Foundation Model Research Projects:** Interns should be given opportunities to work directly with cutting-edge models such as LLaMA 3 and other open-source large language models. Encouraging early-career researchers to participate in tasks like fine-tuning, prompt engineering, and evaluation in domain-specific contexts would greatly accelerate skill development in applied NLP. Clear documentation, shared model checkpoints, and access to compute resources are key to enabling meaningful contributions.
- 2. Foster Collaboration with Institutions:** Collaborate with educational institutions to involve students in research projects, competitions, and workshops. Establish partnerships to offer joint internship programs, research collaborations, and academic exchanges. Engage with faculty members and research centers to tap into expertise and resources, enriching the internship experience and fostering closer ties between academia and industry.
- 3. Provide Mentorship and Support:** Offer mentorship and guidance to interns, pairing them with experienced professionals who can provide advice, feedback, and support. Mentors can help interns navigate challenges, develop new skills, and achieve their goals. Regular check-in meetings, mentorship workshops, and networking opportunities can enhance the mentorship experience.
- 4. Encourage Innovation and Creativity:** Create an environment that encourages interns to explore innovative ideas and contribute to ongoing projects. Empower interns with autonomy and resources to pursue their interests, participate in brainstorming sessions, hackathons, and design challenges to stimulate creativity and inspire new solutions to complex problems.
- 5. Facilitate Networking Opportunities:** Organize networking events and activities to enable interns to connect with professionals from various departments. Plan meet-and-greet sessions, panel discussions, and industry tours where interns can interact with employees, alumni, and industry experts. Establish online networking platforms to facilitate ongoing communication and collaboration.



## CHAPTER 9

### CONCLUSION

In conclusion, these projects tackled two key areas in the evolving landscape of large language models and information retrieval: embedding model comparison for Retrieval-Augmented Generation (RAG) and fine-tuning LLaMA 3 for downstream tasks. Both components represent critical advancements in making large language models more accurate, responsive, and domain-aware.

The first half of the project focused on evaluating embedding models (such as BGE, E5, and GTR) for use in RAG pipelines. Through systematic benchmarking across various corpora, we analyzed how different embeddings impacted retrieval quality using metrics like Recall@k and nDCG. This comparative study provided valuable insights into the trade-offs between model size, latency, and retrieval precision—ultimately guiding optimal model selection for domain-specific applications.

The second half of the project involved fine-tuning LLaMA 3, an open-source large language model, to adapt it for custom use cases. By leveraging techniques such as instruction tuning and prompt engineering, the model was refined to improve performance on specific tasks. This process deepened understanding of model behavior, tokenization effects, and the importance of high-quality training data. The experimentation also revealed the practical challenges and considerations involved in training large models, including resource constraints and hyperparameter tuning.

Collectively, this work demonstrates the power of combining robust retrieval mechanisms with fine-tuned generative models in building intelligent, grounded AI systems. The successful outcomes from both embedding model comparison and LLaMA 3 fine-tuning highlight the potential for more scalable and accurate RAG-based architectures in real-world scenarios.

Looking ahead, further research could explore hybrid retrieval models, unsupervised fine-tuning techniques, or incorporate user feedback loops for continuous improvement. This project has laid a strong foundation in applied NLP and model evaluation, offering valuable technical experience and a deepened appreciation for the intricate dynamics between retrieval and generation in modern AI systems.

## CHAPTER 10

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